The legendary sword of King Arthur, Excalibur, holds such ultimate power that after the death of King Arthur, it has been **broken into three pieces**—jewel, blade and hilt—to be buried underground. After a long period of time, Greg, a descendant of King Arthur who lived an ordinary life as a farmer discovered a map in his home. It was a map of the location where the three pieces of Excalibur were buried. If someone were to put together the three pieces of the Excalibur to forge a complete Excalibur, that person would get to hold ultimate power and become the new king.

The map shows a large historical site with **the length of N km and width of M km**. Greg can move up, right, left and down on the map, and the map symbol indicates the surface area of 1km2 . The map is marked with many symbols which refers to the following, respectively.

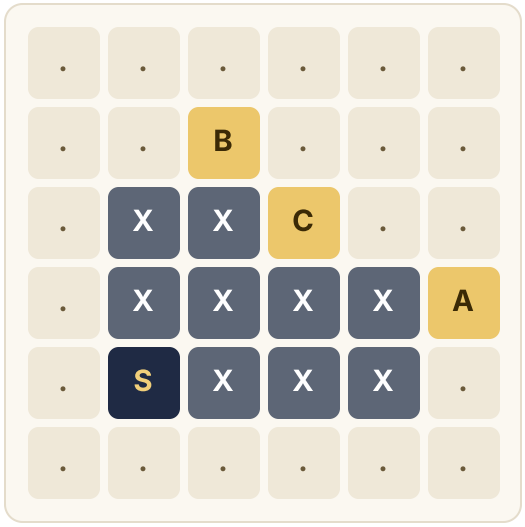
`.` : plain	`A` : jewel of Excalibur	`C` : hilt of Excalibur
`X` : mountain	`B` : sword of Excalibur	`S` : royal palace

Greg was determined to collect A, B and C pieces of Excalibur **to forge a complete Excalibur and become the new king of the royal palace**. As bandits live in the mountains, it is impossible for Greg to pass by. However, if carrying the Excalibur, the bandits will not be able to approach Greg, **enabling him to pass the mountains**. Greg wastes no time to collect the broken pieces of the sword and get to the palace as he wishes to become King as soon as possible.

> When given the length and width of the map, along with the location of Greg in the map and map data, **find the route with shortest distance for Greg to make a complete Excalibur and get to the palace**.

[Example]

N = 6, M = 6, Greg's starting point = (1, 4)



For instance, let's assume that a map with the length of 6 and width of 6 is given. Let's say the location of A is (4, 6), the location of B is (2, 3), the location of C is (3, 4), the location of the palace is (5, 2), and Greg's starting point is (1, 4). Greg will move in the order of B -> C -> A to collect the broken pieces of the Excalibur and after putting together all the pieces into one, he will head towards the palace (S). If he moves in this order, he will be moving 12 times, **which will be the shortest route** he can take.

[Constraints]

1. N and M that indicate the length and width, respectively, are integer numbers from 3 and 100, including 3 and 100.
2. R and K that indicate the row and column of Greg's location in the map, are integer numbers from 1 and N, M, including 1 and N, M.
3. One of each (A, B, C, S) exists in the map by all means.
4. If A, B, and C are collected, the sword Excalibur can be immediately made.
5. There are no odds of Greg failing to complete the Excalibur.
6. Greg's initial location given on the map is a plain.
7. Greg can pass the palace without completing the Excalibur, and nothing will happen.
8. Areas beyond the map are unchartered territories which are forbidden.

[Input]

In the first line, the number of test casts (T) is given, and then T number of test cases will be given. In the first line of each test case, N, M, R, and K that each indicate the map length, width, Greg's location in terms of row and column are given. Across N lines is a map.

[Output]

Print the answers to each test case. Each test case begins with "#x" (x is a test case number, starts from 1) and print the answer of the test case.

[Example of Input and Output]

(Input)

```
2
6 6 1 4
......
..B...
.XXC..
.XXXXA
.SXXX.
......
5 6 2 3
SX...X
XX..AX
.XXB.X
..XX.X
..C..X
```

(Output)

```
#1 12
#2 15
```

(Sample_input.txt output)

```
#1 12                #11 20
#2 15                #12 46
#3 4                 #13 24
#4 4                 #14 49
#5 19                #15 26
#6 47                #16 38
#7 16                #17 33
#8 20                #18 24
#9 16                #19 34
#10 16               #20 48
```